

Unit IV

Introduction to IC Engines & Automobiles

Complete Question Bank — 2M / 3M / 7M Questions + 14 Marks Paper

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Pass hojau bhai bas kal behenchod

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1. 2-Mark Questions

1. Define internal combustion engine.
2. List any two applications of IC engines.
3. Define bore and stroke.
4. What is TDC? What is BDC?
5. Define swept volume.
6. Define clearance volume.
7. Write formula for compression ratio.
8. State Otto cycle processes.
9. State Diesel cycle processes.
10. Define brake power.
11. Define indicated power.
12. What is mean effective pressure?
13. Write brake power formula.
14. What is a carburetor?
15. Function of spark plug.
16. Name two piston materials.
17. Define over-square engine.
18. What is cut-off ratio?
19. Typical CR for petrol engine.
20. Typical CR for diesel engine.
21. Define mechanical efficiency.
22. What is a two-stroke engine?
23. What is a four-stroke engine?
24. What is a ladder frame?
25. What is a monocoque frame?
26. Two active safety features.
27. Two passive safety features.
28. What is a BEV?
29. Name two EV batteries.
30. Define torque.
31. Define RPM.
32. Name two frame cross-sections.
33. Function of piston rings.
34. Name the part converting reciprocating motion to rotation.

2. 3-Mark Questions

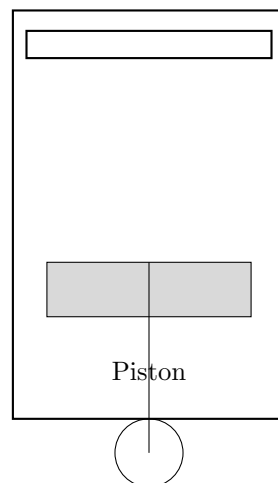
1. Explain bore, stroke and derive swept volume.
2. Define and explain compression ratio.
3. Explain four strokes of SI engine.
4. Working of two-stroke engine.
5. Sketch piston-connecting rod-crank.
6. Functions of carburetor, intake and exhaust manifolds.
7. Explain mean effective pressure.
8. Differentiate indicated power and brake power.
9. Explain Otto cycle and write efficiency formula.
10. Explain Diesel cycle and effect of cut-off ratio.
11. Explain formula $BP = \frac{2\pi NT}{60}$.
12. Draw Otto vs Diesel PV sketch.
13. Explain piston rings and their types.
14. Explain over-square vs under-square.
15. Explain connecting rod and crankshaft.
16. Explain valve timing.
17. Define IMEP and BMEP.
18. Ladder frame vs monocoque frame.
19. Explain ABS.
20. PMSM vs induction motor in EVs.
21. What is a hybrid (PHEV)?
22. Explain mechanical efficiency.
23. Explain knocking basics.
24. Factors affecting Otto efficiency.
25. Explain torque and RPM curves.
26. Mean piston speed significance.
27. Camshaft and valve actuation.
28. Describe EV battery selection factors.
29. SI vs CI ignition.

3. 7-Mark Questions

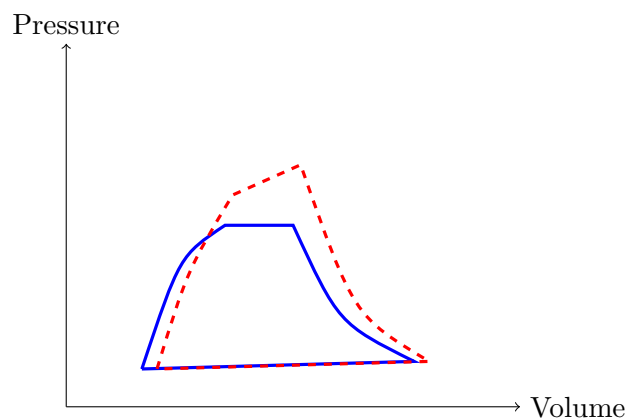
1. Draw four-stroke SI engine and explain all strokes with valve events.
2. Draw Otto cycle PV & TS and derive air-standard efficiency.
3. Derive Diesel cycle efficiency.
4. Solve a compression problem using $pv^\gamma = \text{constant}$.
5. Compare Otto and Diesel in detail.
6. Explain carburetor construction & AFR control.
7. Types of cooling systems.
8. Two-stroke engine with scavenging sketch.
9. Explain IP, BP, FP and show relation $IP = BP + FP$.
10. Explain torque measurement and BP computation.
11. Describe vehicle frames with sketches.
12. Draw chassis layout and explain components.
13. Explain ABS, ESC, airbags.
14. Explain PMSM & BLDC motors.
15. EV batteries — Li-ion, solid state — properties.
16. Derive engine displacement for 4-cylinder engine.
17. Explain IMEP and derive power equation.
18. Explain knocking in detail.
19. Emissions and after-treatment basics.

4. Important Diagrams

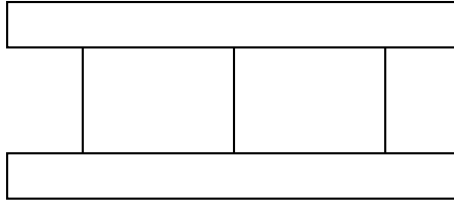
4.1. Four-stroke Engine (Simplified)



4.2. Otto (blue) vs Diesel (red) PV Sketch



4.3. Ladder Frame



5. Model 14-Mark Question Paper

- Q1 (14M)** (a) Draw and explain four-stroke SI engine. (8M)
(b) Given $p_1 = 1$ bar, $p_2 = 11$ bar, $\gamma = 1.4$, compute compression ratio and Otto efficiency. (6M)
- Q2 (14M)** (a) Draw Otto & Diesel PV diagrams. (6M)
(b) Write Diesel efficiency formula and explain effect of cut-off ratio. (8M)
- Q3 (14M)** (a) Explain IP, BP, FP. (6M)
(b) Compute BP for $T = 80$ Nm, $N = 3000$ rpm. (4M)
(c) Explain BMEP significance. (4M)
- Q4 (14M)** Two-stroke engine diagram & scavenging. (7M)
Two-stroke vs four-stroke comparison. (7M)
- Q5 (14M)** (a) Ladder vs monocoque frame. (6M)
(b) ABS, ESC, airbags. (8M)